

Slope-Intercept Form ($y = mx + b$)

Practice problems on slope, y-intercept, and slope-intercept form

Name: _____ Date: _____ Class: _____

1. Identify m and b from the equation

In $y = mx + b$, the number in front of x is the slope; the constant is the y-intercept.

① $y = 2x + 3$ $\rightarrow m =$ _____ $b =$ _____

② $y = -4x + 7$ $\rightarrow m =$ _____ $b =$ _____

③ $y = (1/2)x - 1$ $\rightarrow m =$ _____ $b =$ _____

④ $y = -(2/3)x + 5$ $\rightarrow m =$ _____ $b =$ _____

⑤ $y = x + 6$ $\rightarrow m =$ _____ $b =$ _____

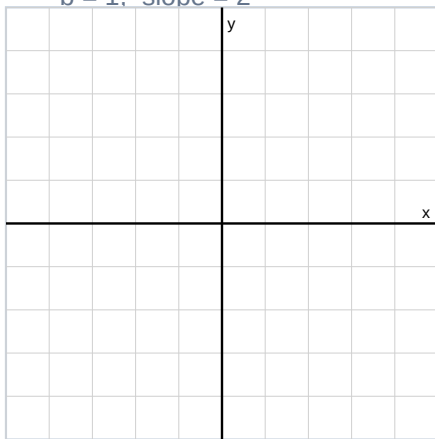
⑥ $y = -3x$ $\rightarrow m =$ _____ $b =$ _____

2. Graph each line — start at b, then use slope

Plot the y-intercept (0, b) first. Then use slope = rise/run to plot a second point.

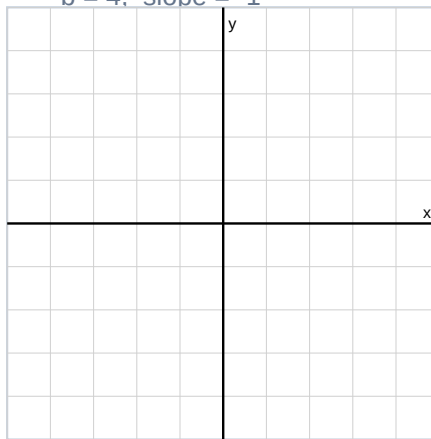
⑦ $y = 2x + 1$

b = 1, slope = 2



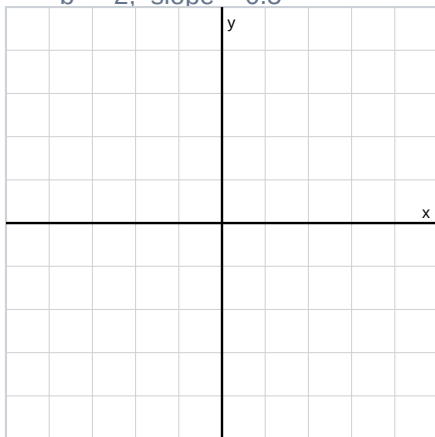
⑧ $y = -x + 4$

b = 4, slope = -1



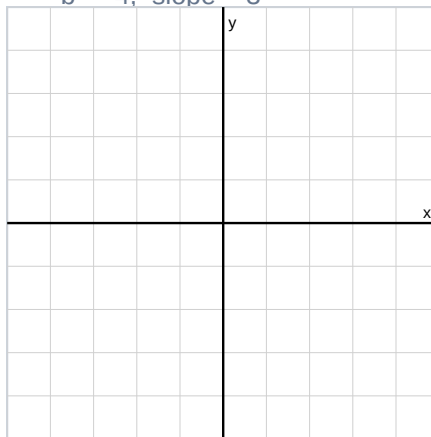
⑨ $y = (1/2)x - 2$

b = -2, slope = 0.5



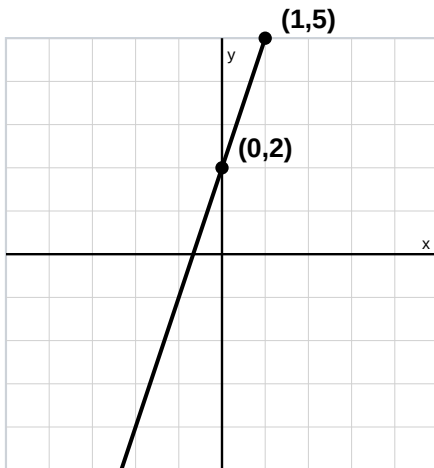
⑩ $y = 3x - 4$

b = -4, slope = 3



3. Write the equation from the graph

Find where the line crosses the y-axis (that's b). Count rise/run for m . Write $y = mx + b$.

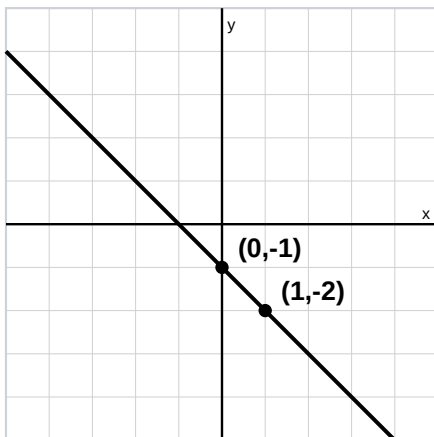


⑪ Read the graph and write:

$$m = \underline{\hspace{2cm}}$$

$$b = \underline{\hspace{2cm}}$$

$$y = \underline{\hspace{3cm}}$$

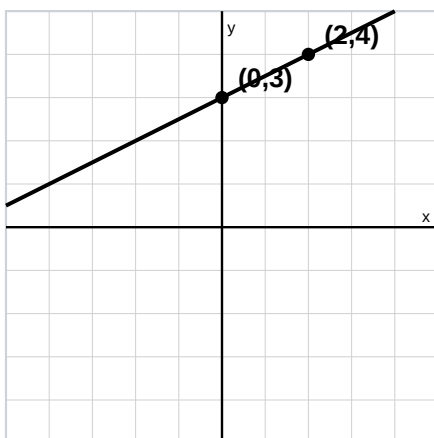


⑫ Read the graph and write:

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$$b = \underline{\hspace{2cm}}$$

$$y = \underline{\hspace{3cm}}$$



⑬ Read the graph and write:

$$m = \underline{\hspace{2cm}}$$

$$b = \underline{\hspace{2cm}}$$

$$y = \underline{\hspace{3cm}}$$

4. Rewrite in slope-intercept form (solve for y)

Get y by itself. Move x-term to the other side, divide if needed.

⑭ $3x + y = 5$ $\rightarrow y =$ _____ $m =$ _____ $b =$ _____

⑮ $2x - y = 4$ $\rightarrow y =$ _____ $m =$ _____ $b =$ _____

⑯ $x + 2y = 6$ $\rightarrow y =$ _____ $m =$ _____ $b =$ _____

⑰ $-4x + 2y = 8$ $\rightarrow y =$ _____ $m =$ _____ $b =$ _____